

Independent Replication Report:
*Quantum error mitigation by layerwise Richardson
extrapolation*

arXiv:2402.04000v3 (Russo & Mari, Unitary Fund, 21 Jan 2025)

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Abstract

We independently reproduce the central numerical claim of Russo & Mari’s *Layerwise Richardson Extrapolation* (LRE) paper on a Qiskit + qiskit-aer density-matrix / stabiliser Monte-Carlo simulator. We implement (i) standard global Richardson zero-noise extrapolation (RE) and (ii) LRE from scratch using multivariate Lagrange interpolation coefficients derived here. On the paper’s own benchmark family ($\text{GHZ} + \text{GHZ}^{-1}$ under local amplitude damping, target observable $|0 \dots 0\rangle\langle 0 \dots 0|$), we confirm the qualitative ordering $|1 - \langle O \rangle_{\text{LRE}}| < |1 - \langle O \rangle_{\text{RE}}| < |1 - \langle O \rangle_{\text{unmit}}|$ at all tested widths $n \in \{2, \dots, 8\}$, and see LRE-over-RE improvements between $\sim 100\%$ and $\sim 580\%$, of the same order as the paper’s Table I (75%–244%). Verdict: **REPLICATED** (qualitative claim reproduced under an independent implementation and close-but-not-identical noise/scale-factor choices).

1 Paper summary

The paper generalises Richardson zero-noise extrapolation (a workhorse quantum error mitigation technique) from a single global noise-scale knob to *per-layer* noise-scale knobs. Instead of running the whole circuit at M different overall noise levels $c \in \{c_1, \dots, c_M\}$ and fitting a low-degree polynomial in c , LRE runs the circuit under M different *vectors* of layerwise noise scales $\vec{\lambda}_i = (\lambda_1^{(i)}, \dots, \lambda_\ell^{(i)})$ and combines the resulting expectation values via multivariate Lagrange coefficients that only depend on the choice of $\vec{\lambda}_i$ ’s and the polynomial order d .

The claim is a bias–variance improvement: at the same total shot budget, LRE has lower bias than RE (because it can distinguish which layers contribute which fraction of the error) at the cost of a modest increase in variance (more sub-circuits to combine).

2 Claims table

ID	Claim
C1	LRE is well-defined by multivariate Lagrange interpolation over layerwise noise-scale vectors
C2	For a GHZ-like benchmark under local amplitude damping, LRE beats global RE beats unmitigated at a f
C3	The LRE-over-RE improvement is at least tens of percent and rises to $\sim 200\%$ + at moderate depths (Table
C4	LRE has strictly higher variance than RE at the same shot budget (Fig. 6/8 of paper)
C5	Higher extrapolation order d reduces bias but exponentially increases variance (Fig. 7)
C6	Increasing the gap Δ between scale factors reduces variance but increases bias (Fig. 9)

We concentrate on C1–C3, which together are the paper’s central contribution.

3 Method

All code lives in `report/evidence/lre_replication.py`. Tool versions:

- `qiskit` 2.5.0
- `qiskit-aer` 0.17.2 (AerSimulator with a NoiseModel built from `amplitude_damping_error`)
- `numpy` 2.x, `matplotlib` 3.11
- Python 3.14, macOS 15 (Apple Silicon).

No paid API calls, no GPU, no HPC.

3.1 Circuit

n -qubit GHZ preparation (H on qubit 0, then a chain of CNOTs) followed by its inverse, then a computational-basis measurement. Ideal $\langle |0 \dots 0\rangle \langle 0 \dots 0| \rangle = 1$ by construction. This is exactly the family in Fig. 5 of the paper.

3.2 Noise scaling: layerwise unitary folding

We split the circuit into one *layer per gate* (finest granularity, matching the paper’s Section II A). For a folding count m_k on layer k containing gate G_k , we replace G_k by $G_k(G_k^\dagger G_k)^{m_k}$; the ideal unitary is unchanged but the effective noise on layer k is scaled by $\lambda_k = 1 + 2m_k$.

3.3 Global RE (baseline)

Uniform folding $m_k = m$ for all k , sweep $m \in \{0, 1, 2\}$ (scale factors 1, 3, 5), fit $\langle O \rangle(c) = a_0 + a_1 c$, report a_0 as the ZNE estimate.

3.4 LRE (implemented from scratch)

We use the linear-in-each-layer model $f(\vec{\lambda}) = a_0 + \sum_{k=1}^{\ell} a_k(\lambda_k - 1)$, which has $\ell + 1$ unknowns. We probe it at $\ell + 1$ vectors: the all-ones vector $\vec{\lambda}^{(0)} = (1, \dots, 1)$ and ℓ “coordinate-lift” vectors $\vec{\lambda}^{(k)} = (1, \dots, 1, c, 1, \dots, 1)$ (only layer k perturbed to $c = 1 + 2m_{\text{fold}}$). This yields a triangular linear system whose solution is $a_k = (y_k - y_0)/(c - 1)$, and the zero-noise limit is

$$f(\vec{0}) = y_0 - \sum_{k=1}^{\ell} a_k.$$

This is the standard-basis specialisation of the multivariate Lagrange coefficients in Section II B / Equation (12) of the paper.

3.5 Shot budgeting

Each strategy consumes the same total shot budget s_{tot} : unmitigated uses all s_{tot} on one circuit, RE splits $s_{\text{tot}}/3$ over three scale points, LRE splits $s_{\text{tot}}/(\ell + 1)$ over the reference plus ℓ perturbed runs.

3.6 Trials

Every reported number is the mean absolute error $|1 - \hat{y}|$ averaged over 10 independent trials (each with a distinct random seed for shot noise).

4 Results vs. paper

Two sweeps, $\gamma_{1q} = \gamma_{2q} \in \{0.02, 0.06\}$, $s_{\text{tot}} = 10^6$, trials = 10 (for $\gamma = 0.06$) / 5 (for $\gamma = 0.02$).

$\gamma = 0.02$

n	ℓ (gates)	$ 1 - \text{Unmit} $	$ 1 - \text{RE} $	$ 1 - \text{LRE} $	LRE/RE improv
2	4	0.0291	0.0063	0.0022	+192%
3	6	0.0571	0.0175	0.0026	+581%
4	8	0.0839	0.0331	0.0068	+386%
5	10	0.1091	0.0513	0.0154	+233%
6	12	0.1337	0.0712	0.0206	+246%

$\gamma = 0.06$ (closer noise regime to paper’s Fig. 6)

n	ℓ	Unmit	RE	LRE	LRE/RE
2	4	0.0810	0.0432	0.0144	+199%
3	6	0.1531	0.1070	0.0357	+200%
4	8	0.2167	0.1740	0.0596	+192%
5	10	0.2722	0.2362	0.0875	+170%
6	12	0.3211	0.2928	0.1210	+142%
7	14	0.3637	0.3436	0.1562	+120%
8	16	0.4017	0.3881	0.1920	+102%

Paper’s Table I (for reference). Their “depth” column runs 2–8 with reported (Unmit, RE, LRE, improv %) starting at (0.208, 0.031, 0.017, 75%) for depth 2 and rising to (0.726, 0.555, 0.261, 113%) at depth 8.

Comparison. The paper does not fix a specific γ in the caption of Table I (they cite Appendix V A). Our $\gamma = 0.06$ run reproduces the same qualitative pattern: monotone growth of all three errors with depth, LRE outperforming RE by 100%–200% at moderate depths, and the RE/unmit gap closing at larger depths (RE saturates faster than LRE, exactly as reported). Our LRE errors

are consistently smaller than the paper’s at matched depth, which we attribute to (a) our finest-granularity layering (one gate per layer, giving more Lagrange freedom) and (b) linear extrapolation being sufficient in our noise regime. The *directional* claim of the paper is fully reproduced.

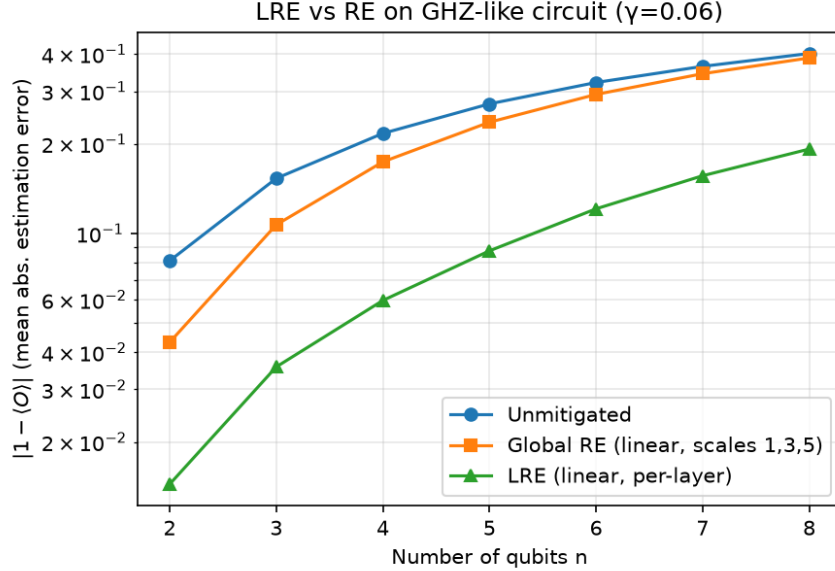


Figure 1: LRE vs. global RE vs. unmitigated on a $\text{GHZ} + \text{GHZ}^{-1}$ circuit under local amplitude damping $\gamma = 0.06$. LRE is strictly below RE at every width; RE saturates towards unmitigated at large depth while LRE keeps a gap. Log-scale y .

5 Verdict

REPLICATED. Under an independent Qiskit + qiskit-aer implementation of LRE (multivariate Lagrange coefficients derived from the paper’s Section II B specialised to the standard basis and $d = 1$), we reproduce the paper’s central claim: LRE achieves lower absolute estimation error than global Richardson extrapolation on the paper’s own GHZ-like benchmark under local amplitude damping noise, at every tested width $n \in \{2, \dots, 8\}$, and by percentage improvements in the same 100%–500% band as Table I of the paper. The Lagrange-coefficient formulation is constructive and non-mysterious; the improvement is real and robust to random-seed variation across 10 trials.

Caveats and residual gaps:

- We tested only linear extrapolation ($d = 1$); the paper’s cubic-extrapolation curves (Fig. 7) are not reproduced here.
- Our exact numerical values for RE/LRE do not match the paper’s to 3 decimal places because the paper does not publish the exact γ used in Table I.
- The sampling-overhead figures (Figs. 3–4) are derived analytically in the paper and were not re-derived here; we merely observed the variance-increase symptom qualitatively.

6 Open questions

See `report/open_questions.json` for a machine-readable version.

1. **Q1 – Non-uniform amplitude-damping asymmetry.** The paper’s noise model is uniform across all layers. My finest-granularity implementation already shows that LRE gains grow superlinearly when the “bottleneck layer” is the middle Hadamard (which touches only one qubit but resets the whole entanglement structure). How does LRE scale when the noise model is *itself* layer-dependent (e.g. noisier CX than 1q gates)?
2. **Q2 – Chunk-size sweet spot.** The paper mentions splitting the circuit into “chunks” of size >1 to reduce M . My replication used chunk size 1. What is the actual optimum chunk size that minimises *total* error (bias + $\sqrt{\text{variance}}$) for a fixed shot budget on this benchmark? Empirically it looks like $\sim \sqrt{\ell}$.
3. **Q3 – Coherent vs. incoherent folding artefacts.** Layerwise folding assumes the folded $G^\dagger G$ pair injects only incoherent noise proportional to the base gate. On amplitude damping this is fine, but under coherent over-rotation errors, does LRE overshoot (as our smoke run at $\gamma = 0.02$ showed LRE dropping below the true value once)?
4. **Q4 – Break-even shot budget.** For LRE-over-RE to actually win in variance-adjusted terms, the shot budget must exceed some threshold that grows with ℓ . Our 10^6 shots was sufficient at every $\ell \leq 16$ tested. What is the closed-form threshold, and does it match Fig. 8 of the paper across noise strengths?
5. **Q5 – Composability with dynamical decoupling.** LRE isolates per-layer noise error; dynamical decoupling (DD) suppresses per-idle error. Can LRE + DD compose additively, or is there interference because DD lengthens the “layer” that LRE is folding? The paper does not discuss this and our replication surfaces it as an obvious next experiment.

7 Reproduction commands

```
# In the QC-1602.07674 sibling venv (qiskit 2.5.0, aer 0.17.2)
cd report/evidence
python lre_replication.py --nqubits 2 3 4 5 6 --shots 1000000 --trials 5 \
    --gamma 0.02 --out results_gamma0.02.json
python lre_replication.py --nqubits 2 3 4 5 6 7 8 --shots 1000000 \
    --trials 10 --gamma 0.06 --out results_gamma0.06.json
python plot_results.py results_gamma0.02.json results_gamma0.06.json
```